TANMAY RAUT

ERP & Business Systems Lead | Manufacturing Process Design | Requirements, Implementation, Adoption

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SUMMARY

Business systems owner for a high-mix, low-volume manufacturer: sole accountable owner of the ERP across eight functions (manufacturing, inventory, quality, procurement, planning, sales, shipping, accounting) and 500+ active SKUs. Five years in manufacturing operations and enterprise systems. I lead current-state assessment, future-state process design, and requirements definition, then stay accountable through configuration, UAT, go-live, and adoption. I do not own hands-on system build; developers and AI-assisted development tools build to my specifications while I own the design, the test, and the outcome. Lean Six Sigma Green Belt. MS Mechanical Engineering, Arizona State University.

CORE COMPETENCIES

Process design: current-state / future-state mapping, swimlane and BPMN process maps, gap and root-cause analysis, process standardization across functions

Requirements and delivery: user requirements specifications, fit-gap analysis, standard-vs-custom decisions, functional specs for developers, test scripts and UAT traceability, hypercare

Manufacturing systems: BOM and routing architecture, work orders and work centers, MRP and reorder rules, lot/serial traceability, quality inspection and RMA workflows, capacity and lead-time planning

Finance integration: manufacturing cost accounting, inventory valuation (AVCO/standard), WIP and overhead absorption, three-way match, deferred revenue, sales tax and exemption handling

Platforms: Odoo 17/19 (expert, 3 yrs sole owner), SAP S/4HANA process alignment, Shopify, Stripe, Avalara, ShipStation, Slack, Ramp; SQL reporting (AI-assisted authoring, GL-validated); Excel/Power Query, Tableau

Change management: stakeholder facilitation without formal authority, role-based training, SOPs and governance documentation, adoption tracking; Lean, Kaizen, 5S, VSM, SPC, FMEA

PROFESSIONAL EXPERIENCE

ERP Systems Engineer II (Business Systems Owner) | Heckler Design Inc, Phoenix, AZ **Apr 2024 – Present**

Sole owner of ERP strategy, design, governance, and rollout for a design-and-manufacturing company. Partner to the controller, production, purchasing, planning, and customer service leads.

- Led current-state assessment of manufacturing costing and found 3D-printed components valued at ~60% of true cost; traced to four configuration defects across 1,840 routing operations and 215 open work orders; wrote the correction specification, validated against the general ledger with the controller, deployed with zero production interruption, and documented the monthly absorption procedure.
- Designed future-state quality and vendor-return process (15 defect types, 6 disposition paths, automated draft credit notes, vendor self-service portal); authored requirements, directed the AI-assisted build, ran UAT with operations and accounting, and drove adoption across three departments and external vendors.
- Integrated manufacturing cost accounting end to end: work-order labor, machine time, material, and overhead to WIP and finished goods, with three-way match on procurement and scrap tracking, giving finance full cost traceability from the floor to the financial statements.
- Specified and delivered operational KPI dashboards (cycle time, rework, yield, backlog, throughput) used by leadership weekly; SQL authored with AI assistance and reconciled to ledger balances before release.
- Designed the variant-based product and BOM structure for a configure-to-order product line so web-configured orders flow to the correct manufacturing order, components, and cost automatically.
- Own cross-platform integrations (Shopify, Stripe, Avalara, ShipStation, Slack, Ramp): requirements, vendor and developer management, testing, and production support; designed sales-tax and exemption-certificate handling rules for returns and fees.
- Led an AI-to-ERP connector initiative that lets any staff member query live production data in plain language; defined scope and guardrails, validated outputs, and rolled it out company-wide.

ERP Systems Engineer | Heckler Design Inc, Phoenix, AZ **Oct 2023 – Mar 2024**

- Led the full Odoo 16 to 17 upgrade across all departments: coordinated an offshore development team, owned end-to-end test plans and UAT, trained every department, and delivered a 65-hour cutover with zero data loss.
- Launched MRP with automated reordering and demand-triggered procurement, replacing spreadsheet-based material planning and cutting raw-material stockouts.

- Implemented the accounting module as the company's system of record: AP, inventory valuation, manufacturing cost accounting, bank reconciliation, invoicing, and credit notes.
- Designed work-order generation and lead-time tracking automation; delivered through AI-assisted configuration with full UAT.

Manufacturing Engineer | United Foods, Phoenix, AZ **Jul 2023 – Oct 2023**

- Led continuous-improvement projects on high-volume food production lines: time studies, standardized work instructions, line balancing, and preventive-maintenance scheduling to raise hourly output and cut downtime.

Production Supervisor, Industrial Fabrication | Vats and Vessels Ltd, Mumbai, India **Jan 2020 – Dec 2022**

- Supervised 30+ welders, machinists, and inspectors fabricating pressure vessels under ASME Section VIII and ISO 9001; developed BOMs, routing sheets, and operator logs; applied 5S and Kaizen to lift first-pass yield; zero reportable safety incidents over 12+ months.

KEY INITIATIVES AND MEASURABLE RESULTS

Finance integration and controls

- Inventory valuation rebuild: when a platform upgrade removed the valuation-layer history, designed a replacement aging and valuation report from on-hand quant data so finance kept month-end inventory reporting without a gap.
- Stopped an automated bank-reconciliation rule that was mis-matching customer wires to unrelated vendor entries; traced the exposure, corrected mislabeled ledger lines, and retired the rule permanently.
- Sales-tax platform migration (Avalara to Stripe Tax) for e-commerce orders: wrote the business rules for restocking fees, return taxation, and exemption certificates; validated live on a test environment before any production change.
- Fixed a unit-of-measure defect on 17 BOM lines that had overstated component cost for two product families; coordinated the five-figure inventory write-down with the controller and added a nightly job so shipped-order margins stay consistent.
- Found and closed a purchase-order defect that wiped unit prices on quantity edits (one PO understated by a five-figure amount); shipped a guard, corrected COGS through three journal entries, and documented the procedure.

Manufacturing, planning, and capacity

- Specified a lead-time calculator and operations issues board used by purchasing and planning daily: BOM gaps, missing reorder rules, and MO timing surfaced automatically instead of by spreadsheet.
- Specified a live 3D-print capacity planner (18/4/22 printer pools, 8-hour shift model fed from gate actuals) that replaced a frozen HTML dashboard with a real-time view inside the ERP.
- Corrected manufacturing-planning calendars, holidays, and 24-hour printer availability so scheduled dates matched the floor; removed a procurement rule that had spawned 79 phantom delivery orders.

Order-to-cash, customer, and channel systems

- Owned the ERP side of a customer-portal RMA lifecycle: customer request to Odoo RMA, receipt validation, automatic accounting hand-off, and two-way message sync; shipped to production with the web team.
- Removed a signature gate that blocked auto-confirmation on 650+ paid web orders; redesigned prepayment handling to deferred revenue per the controller's requirements.
- Defined dealer and distributor classification on the customer master (mutually exclusive, inherited to contacts), ran a duplicate-email audit across 6,150 contact cards, and reassigned 780 accounts to the correct sales owners with full rollback files.
- Led customer master-data cleanup: archived 12,950 dead address cards, fixed a 2022 defect that created every new contact as an invoice address, and documented the channel-routing model shared with the website.

Governance, data quality, and people systems

- Established a Test-then-Production release discipline with drift checks between environments; moved point-and-click customizations into versioned modules to eliminate upgrade breakage.

WORK SAMPLES AND PORTFOLIO

- tanmay-portfolio-pied.vercel.app: case studies (Odoo 16 to 17 cutover, vendor portal, QC/RMA module) and live demos built from sanitized production data. Requirements specification URS-MFG-001 (manufacturing cost integrity) available on request.

EDUCATION AND CERTIFICATIONS

Master of Science, Mechanical Engineering — Arizona State University, Tempe, AZ (GPA 3.8)

Bachelor of Technology, Aeronautical Engineering (Minor: Mechanical) — Manipal Institute of Technology, India (GPA 3.5)

Lean Six Sigma Green Belt (IIBA) | SAP S/4HANA Essentials | Supply Chain Operations (Rutgers) | Root Cause Analysis & Corrective Action | Tableau for Data Visualization | Odoo ERP functional certification (in progress)